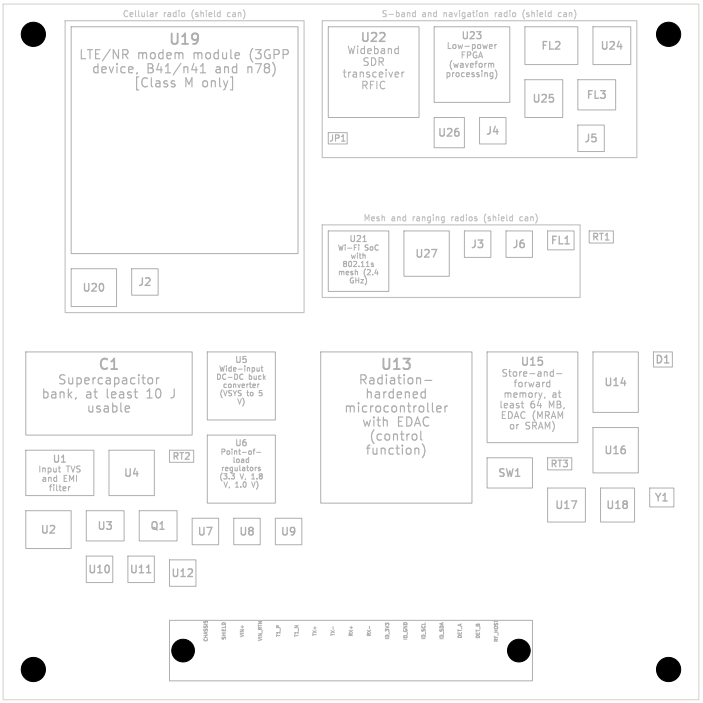


PART TYPES (generic placeholders, no part numbers)

J1	Host connector, blind-mate, shuttered (PBS-HW-SCM-01 Section 5.2)
MH1–4	Mounting hole, chassis bonded
U1	Input TVS and EMI filter
U2	Hot-swap controller with inrush limit
U3	Swap-detect conditioning (comparator)
U4	Hold-up controller (ideal diode and supercapacitor charger)
C1	Supercapacitor bank, at least 10 J usable
U5	Wide-input DC-DC buck converter (VSYS to 5 V)
U6	Point-of-load regulators (3.3 V, 1.8 V, 1.0 V)
U7	Voltage supervisor with watchdog
U8	Current-limited load switch (eFuse), latch-up protection: cellular
U9	Current-limited load switch (eFuse), latch-up protection: mesh
U10	Current-limited load switch (eFuse), latch-up protection: S-band and navigation
U11	Current-limited load switch (eFuse), latch-up protection: ranging
U12	Current-limited load switch (eFuse), latch-up protection: S-band amplifier
U13	Radiation-hardened microcontroller with EDAC (control function)
U14	Radiation-tolerant NOR flash (two firmware images)
U15	Store-and-forward memory, at least 64 MB, EDAC (MRAM or SRAM)
U16	Single-pair Ethernet PHY (100BASE-T1)
U17	RS-422 transceiver
U18	Hot-swap I2C bus buffer with tag supply switch
Y1	TCXO reference oscillator
D1	Ready indicator LED
SW1	Release-request switch (latch handle)
U19	LTE/NR modem module (3GPP device, B41/n41 and n78) [Class M only]
U20	eSIM (MFF2) [Class M only]
J2	Coaxial antenna port: cellular
U21	Wi-Fi SoC with 802.11s mesh (2.4 GHz)
FL1	2.4 GHz band-pass filter (2400–2483.5 MHz)
J3	Coaxial antenna port: 2.4 GHz mesh
U22	Wideband SDR transceiver RFIC
U23	Low-power FPGA (waveform processing)
U24	S-band power amplifier, 2 W [SCM-L only]
JP1	RF bypass link (SCM-S: fitted in place of the amplifier) [SCM-S only]
U25	S-band front end: T/R switch and LNA
FL2	S-band duplexer (2025–2110 / 2200–2290 MHz)
U26	RF antenna selector switch (lid antenna or host RF port)
J4	Coaxial antenna port: S-band lid patch
FL3	Navigation band SAW filter and LNA (2483.5–2500 MHz)
J5	Coaxial antenna port: navigation patch
U27	UWB ranging transceiver (optional, off by default) [Optional]
J6	Coaxial antenna port: UWB
Q1	Heater driver, low-side MOSFET
H1	Polyimide film heater, zone A (radios) (back)
H2	Polyimide film heater, zone B (power and control) (back)
RT1	Platinum RTD temperature sensor (radio zone)
RT2	Platinum RTD temperature sensor (power zone)
RT3	Platinum RTD temperature sensor (control zone)

Generated by PBS-HW-LIB/kicad/tools/generate.py
Placeholders are not routed; nets show the required interconnect
PBS-HW-SCM-01 open reference design; generic placeholders, no part numbers
Pale Blue Systems Foundation

Sheet:
File: pbs-scm-core.kicad_pcb

Title: PBS-HW-SCM-01 Surface Communication Module, core board

Size: A4	Date: 2026-10-08	**Rev: 0.2**
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